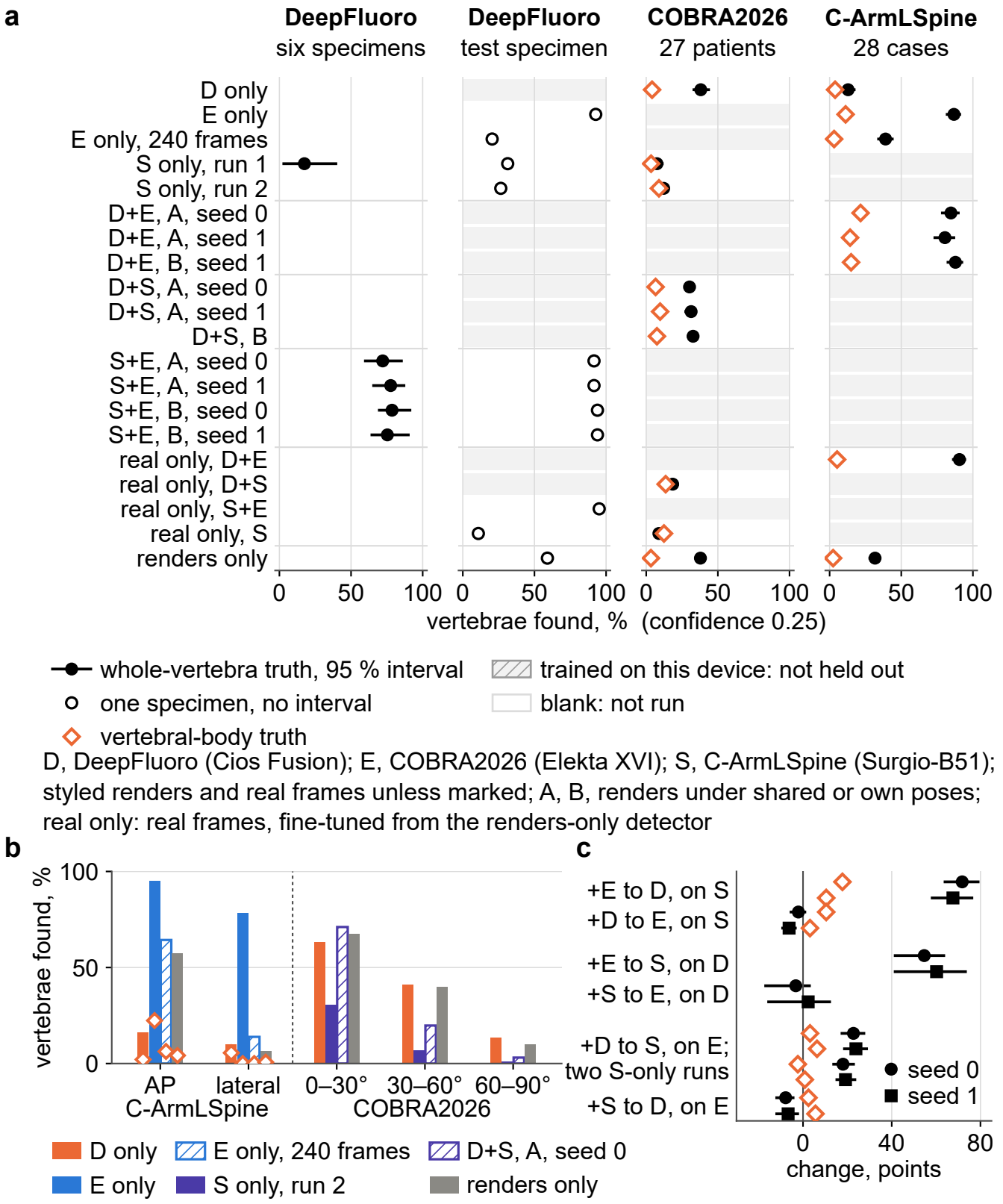

a
DeepFluoro six specimens
DeepFluoro test specimen
COBRA2026 27 patients
C-ArmLSpine 28 cases
D only
E only
E only, 240 frames
S only, run 1
S only, run 2
D+E, A, seed 0
D+E, A, seed 1
D+E, B, seed 1
D+S, A, seed 0
D+S, A, seed 1
D+S, B
S+E, A, seed 0
S+E, A, seed 1
S+E, B, seed 0
S+E, B, seed 1
real only, D+E
real only, D+S
real only, S+E
real only, S
renders only
0 50 100
vertebrae found, % (confidence 0.25)
whole-vertebra truth, 95 % interval
one specimen, no interval
vertebral-body truth
trained on this device: not held out
blank: not run
D, DeepFluoro (Cios Fusion); E, COBRA2026 (Elekta XVI); S, C-ArmLSpine (Surgio-B51); styled renders and real frames unless marked; A, B, renders under shared or own poses; real only: real frames, fine-tuned from the renders-only detector
b
vertebrae found, %
100
50
0
AP
lateral
C-ArmLSpine
0–30°
30–60°
60–90°
COBRA2026
D only
E only
E only, 240 frames
S only, run 2
D+S, A, seed 0
renders only
c
+E to D, on S
+D to E, on S
+E to S, on D
+S to E, on D
+D to S, on E; two S-only runs
+S to D, on E
seed 0
seed 1
0 40 80
change, points